Joshua Rosenberg, Ph.D.

Curriculum Vitae

Contact Information

Haslam Family Professor & Associate Professor, STEM Education
Department of Theory and Practice in Teacher Education
The University of Tennessee, Knoxville
513 Claxton, 1122 Volunteer Blvd., Knoxville, TN 37996
jrosenb8@utk.edu
865-974-5973

Scholarly Interests

- Educational data science
- Educational technologies
- Science teacher education

Citation Metrics

Source: Google Scholar (updated 2026-06-15)

- **Total citations:** 7587 — how many times my work has been referenced by other researchers
- **h-index:** 37 — I have 37 papers each cited at least 37 times
- **i10-index:** 73 — how many of my papers have been cited at least 10 times

Research Program

My research program spans two interconnected lines of work in educational data science: using new data sources and computational methods to describe and question educational systems, and weaving data science into teaching and learning with a focus on science education.

In the first line of work, I study educational systems using computational approaches and novel data sources. This includes research on educators' professional learning through social media, critical investigations of how K–12 institutions' social media practices compromise student privacy, and methodological work that integrates machine learning with qualitative analysis.

In the second line of work, I develop and study approaches for data science learning in K–12 settings. This includes designing instruction for statistical reasoning—especially Bayesian thinking—in science classrooms; building tools and curricula that make data science accessible to newcomers; and contributing to the development of educational data science as a field through widely used open-source software, a co-authored book (*Data Science in Education Using R, Routledge*, 2020), and a synthesis of what is known about K–12 data science learning.

Across both lines, I program in R and Python, use advanced statistical methods (especially multilevel models), develop open-source tools, and center partnerships with teachers. My current and near-future work extends these efforts into multimodal learning analytics, community-based data science learning, and the roles of generative AI in education.

Education

Degrees

2018, PhD, Educational Psychology & Educational Technology
Michigan State University

2012, MA, Education
Michigan State University

2010, BS, Biology
University of North Carolina, Asheville

Additional Qualifications

2016, Graduate Certificate, Science Education
Michigan State University

2010, Educator's License - Science and Biology, Teacher Licensure Program
University of North Carolina, Asheville

Professional Experience

2025-present, Haslam Family Professor
University of Tennessee, Knoxville

2026-present, Co-Director (with Dr. Emily Holtz), AI, Data, and Educational Partnerships at the University of Tennessee (ADEPT) Center
University of Tennessee, Knoxville

2023-present, Associate Professor, STEM Education
University of Tennessee, Knoxville

2018-2023, Assistant Professor, STEM Education
University of Tennessee, Knoxville

2012-2018, Graduate Research and Teaching Assistant
Michigan State University

2010-2012, Science Teacher (Biology and Earth Science)
Shelby High School, Shelby, NC

2009-2010, Science Teacher Intern (Biology and Chemistry)
C.D. Owen High School, Swannanoa, NC

Research Affiliations

Faculty Affiliate, Institute for Intelligent Systems, University of Memphis (2026-present)

Steering Committee Member and Affiliated Researcher, Center for the Dynamics of Cultural Complexity (DySoC), University of Tennessee, Knoxville (2024-present)

Affiliated Researcher, AmplifyLearn.AI, University of Washington (2024-present)

Fellowship Roles

Faculty Fellow (Data Science), College of Emerging and Collaborative Studies, University of Tennessee, Knoxville (2023-2025)

Faculty Fellow, Center for Enhancing Education in Mathematics and Sciences, University of Tennessee, Knoxville (2019-present)

Faculty Fellow, Education Research & Opportunity Center, University of Tennessee, Knoxville (2023-2025)

Awards

2025, College of Emerging and Collaborative Studies, Champion of the Year (External Member), University of Tennessee, Knoxville

2024, Professional Promise in Research and Creative Achievement award, University of Tennessee, Knoxville

2024, Lura Odland Excellence award, College of Education, Health, and Human Sciences, University of Tennessee, Knoxville

2023-2024, Expanding Horizons Mid-Career Faculty Development Program, University of Tennessee, Knoxville

2023, GDCP (German Society for the Teaching of Chemistry and Physics) Best Paper Award

2023, NSF CAREER awardee

2023, Faculty Mentor, SEC Emerging Scholars Program Postdoctoral Researcher (Maryrose Weatherton)

2023, Research Worth Reading Award, Research in Artificial Intelligence in Science Education Research Interest Group, National Association for Research in Science Teaching

2023, Outstanding Graduate Research Mentor Award, Graduate Student Senate, University of Tennessee, Knoxville

2022, Early Career Award, Technology as an Agent of Change in Teaching and Learning (TACTL) Special Interest Group (SIG), American Educational Research Association (AERA)

2021, Best Poster Award, Fourteenth International Conference on Educational Data Mining

2021-2022, Open Educational Resources (OER) Research Fellow, William and Flora Hewlett Foundation

2021, Louie M. & Betty M. Phillips Faculty Support in Education Award, University of Tennessee, Knoxville (UTK)

2021, Mentor, Summer Undergraduate Research Internship Program, Office of Undergraduate Research, UTK

2020, Research Assistant Award, Office of Undergraduate Research, UTK

2020, Southeastern Conference (SEC) Visiting Faculty Travel Grant Program (Host: Annelise Russell, Martin School of Public Policy, University of Kentucky)

2016, Best Paper Award, Technological Pedagogical Content Knowledge SIG, Society for Information Technology and Teacher Education International Conference

2014, Outstanding Paper Award, Society for Information Technology and Teacher Education International Conference

Grants

$9,245,867 as Principal Investigator (PI) or Co-PI

PI or Co-PI

2024-2028, PI, *Collaborative Research: Learning How to Help Middle Grades Science Teachers Integrate Data Exploration and Sensemaking in the Classroom* ($2,997,783.00*;* with lead PI Jonathan Griffith, CU, Boulder; UTK sub-contract: $99,296). NSF. NSF Grant No. 2405912.

2024-2025, Investigator, *AI for Education: Developing Curricula, Planning for External Funding, and Connecting Faculty* ($21,000; with *PI* Louis Rocconi), AI Tennessee Initiative, University of Tennessee, Knoxville.

2023-2028, PI, *CAREER: Creatively Reimagining Engagements with Data in Biology Learning Environments*, ($846,612), NSF. NSF Grant No. 2239152.

2022-2025, Co-PI, *Computer Science for Appalachia: Expanding a research-practice partnership to integrate computer science and literacy in rural East Tennessee schools*, ($999,980; with *PI* Lynn Hodge). NSF. NSF Grant No. 2219418.

2022-2025, Co-PI, *Quantifying the robustness of causal inferences: Extensions and applications.* ($899,319.13; *PI*: Kenneth Frank, Michigan State University; UTK subcontract: $105,727). Institute of Education Sciences.

2022-2024, Co-PI, *Broadening participation in introductory computer science: investigating self-assessment practices for increasing student learning and self-efficacy in two institutional contexts* ($299,836; with *PI* Alex Lishinski). NSF. NSF Grant No. 2215245

2022-2023, Co-PI, *Launching a Micro-credential in Educational Data Analytics* ($10,000; with *Co-PI* Louis Rocconi). University of Tennessee, Knoxville's College of Education, Health, and Human Sciences Strategic Investment Program.

2021-2022, PI, *Not only for scientists and engineers: Advancing Bayesian methods for pre-collegiate learners* ($1,991), Supplemental funding to NSF Grant No. 193770 (Dear Colleague Letter: Research Collaboration Opportunity in Europe for NSF Awardees). National Science Foundation.

2020-2021, Co-PI, *Propelling teacher professional development through FAAST feedback on student epistemic views* ($15,000; *PI*: Christina Krist, University of Illinois Urbana-Champaign). Technology Innovations in Educational Research and Design Pilot Projects Program.

2019-2022, Co-PI, *Advancing computational grounded theory for audiovisual data from STEM classrooms* ($1,313,855; *PI*: Christina Krist, University of Illinois Urbana-Champaign; UTK subcontract: $101,469). NSF. https://tca2.education.illinois.edu/ (NSF Grant No. 1920796)

2019-2021, PI, *Understanding the development of interest in computer science: An experience sampling approach* ($346,688). National Science Foundation [NSF]. http://picsul.utk.edu/ (NSF Grant No. 1937700)

2019-2021, Co-PI, *CS for Appalachia: A research-practice partnership for integrating computer science into East Tennessee schools* ($252,453; *PI*: Lynn Hodge, UTK). NSF. (NSF Grant No. 1923509)

2019-2020, PI, *Planting the seeds for computer science education in East Tennessee through a research-practice partnership* ($13,200). Community Engaged Research Seed Program, UTK.

2018-2020, Co-PI, *Exploring how beginning elementary mathematics teachers seek out resources through social media* ($8,820; *PI*: Stephen Aguilar). Herman & Rasiej K-5 Mathematics Initiative, University of Southern California.

Senior Personnel/Investigator

$13,803,319 as Senior Personnel/Investigator

2025-2030, Senior Personnel, Picturing Possibilities and Envisioning Selves; PiPES3 (PI: Erin Hardin), National Institutes of Health, $1,331,086

2025, Senior Personnel (assumed PI August 2025), Building a Sustainable Future: Examining Community College Student Enrollment and Completions in Green Sector Occupations, JP Morgan Chase. https://cehhs.utk.edu/ero/completions-in-green-sector-occupations/

2024-2029, Senior Personnel, AmplifyGAIN: Generative AI for Transformative Learning. University of Washington (PI: Min Sun). $9,999,976. https://www.amplifylearn.ai/amplifygain-generative-ai-for-transformative-learning/

2020-2025, Co-I, *Imagining possibilities in post-secondary education and STEMM in rural Appalachia* ($1,208,563), National Institutes of Health.

2020-2023, Senior Personnel, *Learning analytics in STEM education research institute* ($933,150; *PI*, Shaun Kellogg, North Carolina State University; UTK subcontract: $62,870. National Science Foundation (NSF), NSF Grant No. 2025090

2019-2022, Senior Personnel, *Medical entomology and geospatial analyses: Bringing innovation to teacher education and surveillance studies* ($149,611; *PI*: Rebecca Trout Fryxell). United States Department of Agriculture - Agriculture and Food Research Initiative. (USDA Grant No. 2019-68010-29119) https://www.megabitess.org/

Consultant and Advisory Board Member Roles

2024-2026, Advisory Board Member, *A Multipronged Approach to Small-Teaching Interventions for Reducing Academic Procrastination: A Randomized Control Study via Terracotta,* NCER grant no. R305N240063. Miyake, A. (PI), Bernacki, M., Kane, M., & Snyder, H. (Co-PIs).

2023-2026, Data Science Assessment Consultant, *NSF AISL: Integrating Research and Practice: Reimagining Youth Community Science through Make-and-Take Data Sensing Kits*, NSF grant no. 2314089. Blikstein, P. & Perez, S. (PIs)

2023-2026, Consultant, *WTG: Diffusion of Research on Supporting Mathematics Achievement for Youth with Disabilities through Twitter Translational Visual Abstracts*, Jessica Rodrigues (PI, University of Missouri), National Science Foundation, https://www.nsf.gov/awardsearch/showAward?AWD_ID=2244734&HistoricalAwards=false

2023-2025, Advisory Board Member, *Integration of Computer-Assisted Methods and Human Interactions to Understand Lesson Plan Quality and Teaching to Advance Middle-Grade Mathematics Instruction.* Min Sun (PI). NSF Grant No. 2300291.

2023-2025, Advisor, *Sloan: Aligning Graduate Education and Workforce Opportunities: A Robust, Equity-Focused Landscape Scan of Computing Terminology.* Wofford, A., Perez-Felkner, L., & Staudt Willet, K. B. (PIs). Funded by Alfred P. Sloan Foundation.

2023-2025, Consultant, *Forschen mit epidemischer Unsicherheit Lernen (Project FEUL)* (Learning to do investigations with epistemic uncertainty), Marcus Kubsch (PI; IPN - Leibniz Institute, Kiel, Germany), Joachim Herz Foundation

2022-2027, Advisory Board Member, *Collaborative Research: Enhancing Data Science and Statistics Teacher Education–Transforming and Building Community*, Hollylynne Lee (PI; North Carolina State University), National Science Foundation. https://www.nsf.gov/awardsearch/showAward?AWD_ID=2141727&HistoricalAwards=false

2022-2025, Advisory Board Member, *A Learning Ecosystem for Teaching High School Students Machine Learning Concepts and Data Science Skills in Healthcare and Medicine*, Kathryn Eller (PI; East Bay Educational Collaborative), National Science Foundation. https://www.nsf.gov/awardsearch/showAward?AWD_ID=2148451&HistoricalAwards=false

2022-2025, Advisory Board Member, *Mathematizing, Visualizing, and Power (MVP): Appalachian youth becoming data artists for community learning*, Lynn Hodge (PI), National Science Foundation, https://www.nsf.gov/awardsearch/showAward?AWD_ID=2215004&HistoricalAwards=false

2022-2024, Participating Faculty, *Bayesian Thinking in STEM*, (NSF Grant No. 2215879), https://www.stat.uci.edu/bayes-bats/

2021-2026, Advisory Board Member, *SEER Research Network for Digital Learning Platforms*, Jeremy Roschelle (PI; Digital Promise), Institute for Education Sciences, Digital Learning Platforms to Enable Efficient Education Research Network

2021-2022, Lead STEM Education Consultant,*Supporting Students' Meaningful Engagement With Data From Small Orbiting Satellites*, Art Palisoc (PI), STEM Ed LLC, NSF Phase I SBIR, *STEM Experiments and Games in Low Earth Orbit – Making STEM Learning Fun*

Publications

\+ Denotes a collaboration with a mentee who is a graduate student
^ Denotes a collaboration with a mentee who is an undergraduate student

Scholarly Books (2)

Wang, W., Akcaoglu, M., Rosenberg, J. M., & Kellogg, S. (under contract, anticipated publication late 2026). *Computational social science cookbook with R: A practical guide* (a volume in the *Chapman & Hall/CRC Big Data Series*). CRC Press.

Estrellado, R. A., Freer, E. A., Mostipak, J., Rosenberg, J. M., & Velásquez, I. C. (2020). *Data science in education using R*. Routledge. *Note.* All authors contributed equally. http://www.datascienceineducation.com/

Book for General Audience (1)

Rosenberg, K., & Rosenberg, J.M. (in press, anticipated release winter 2026-2027). Hiking Knoxville: Family Friendly Adventures from the City to the Smokies. UT Press. https://familyhikesaroundknox.com/

Articles Under Review or in Press

Lopez, F., Darriet, C., Santibanez, L., Cervantes-Soon, C., Frankenberg, E., Asson, S., Delcid, G., Serrano, E., & Rosenberg, J.M (revise and resubmit). Enrollment patterns and online portrayals of dual language programs in California: Obstacles and promising efforts for equitable access. *American Educational Research Journal.*

Staudt-Willet, B., Na. H., & Rosenberg, J. M. (in preparation/revising). Understanding the work of educational data scientists: A mixed methods exploration.

Articles Published in Refereed Journals (66)

Robertson, J. R., Logan, M. W., Rosenberg, J. M., & Lombardi, D. (2026). Profiles of scientific thinking. *Journal of Educational Psychology, 118*(5), 829-851. https://doi.org/10.1037/edu0001011

Rosenberg, J. M., Kubsch, M., Sorge, S., Tautz, S., Pritchard, C., Garner, A. V., & Xu, Z. (2026). Embracing Uncertainty in Science: The ABCs of Data and the Confidence Updater. Science Scope, 49(1), 44-50. https://www.tandfonline.com/doi/full/10.1080/08872376.2025.2601357

Fischer, M., Pritchard, C., Xu, Z., & Rosenberg, J. (2025). Finding Your Way into Data Science Education as a Science Teacher. *The Science Teacher, 92*(6), 49-55.

Rosenberg, J. M., Veletsianos, G., Romero-Hall, E., & Allen, E. (2025). The Accessibility of Published Research in AERA Journals, 2010–2022. Teachers College Record, 127(8), 110-118.

Borchers, C., Wang, Y., Hodge, E. M., & Rosenberg, J. M. (2025). Decoding Sentiment Signals: Lessons From the Political Reception of the Common Core and Next Generation Science Standards. *Educational Researcher*. https://doi.org/10.3102/0013189X25133620

Dogucu, M., Kazak, S., & Rosenberg, J. M. (2025). The Design and Implementation of a Bayesian Data Analysis Lesson for Pre-Service Mathematics and Science Teachers. *Journal of Statistics and Data Science Education, 33*(2), 177-188.

Borchers, C., Darriet, C., Rosenberg, J. M., & López, F. (2025). Dual Intent in Dual-language Programs: Internet Data Mining of School District Communications. TechTrends, 69(2), 400-412. https://link.springer.com/article/10.1007/s11528-025-01049-1

Burchfield, M. A., Rosenberg, J. M., & Stegenga, S. M. (2025). Informed or outdated consent? An investigation into the media release policies of school districts in the United States. *Journal of Research on Technology in Education, 57*(4), 766-783.

Beymer, P. N., Schell, M. J., Alberts, K. M., Phun, V., Rosenberg, J. M., & Schmidt, J. A. (advance online publication). Students' situational engagement profiles in formal and informal

science learning environments. *Journal of Research in Science Teaching.* https://onlinelibrary.wiley.com/doi/full/10.1002/tea.22017

+Pritchard, C., +Borchers, C., Rosenberg, J. M., Fox, A. K., & Stegenga, S. M. (2024). The datafication of student information on X (Twitter). *Computers and Education Open, 7*, 1-10. https://www.sciencedirect.com/science/article/pii/S2666557324000375

Rosenberg, J., & Jones, R. S. (2024). Data science learning in grades K–12: Synthesizing research across divides. *Harvard Data Science Review*, *6*(3), 1-30. https://doi.org/10.1162/99608f92.b1233596

Fütterer, T., Omarchevska, Y., Rosenberg, J. M., & Fischer, C. (2024). How do teachers collaborate in informal professional learning activities? An Epistemic Network Analysis. *Journal of Science Education and Technology*, 1-15. https://link.springer.com/article/10.1007/s10956-024-10122-y%3E

+Narvaiz, S., Lin, Q., Rosenberg, J. M., Frank, K. A., Maroulis, S. J., Wang, W., & Xu, R. (2024). konfound: An R Sensitivity Analysis Package to Quantify the Robustness of Causal Inferences. *Journal of Open Source Software, 9*(95), 1-6. https://joss.theoj.org/papers/10.21105/joss.05779.pdf

Carpenter, J. P., Rosenberg, J. M., Kessler, A., Romero-Hall, E., & Fischer, C. (2024). The importance of context in teacher educators' professional digital competence. *Teachers and Teaching*, 1-17. https://www.tandfonline.com/doi/full/10.1080/13540602.2024.2320155

Ranellucci, J., & Rosenberg, J. M. (2023). Students' interest, engagement, and achievement in online high school science courses. *Educational Psychology*, 1-19. https://doi.org/10.1080/01443410.2023.2299703

Staudt Willet, K.B., Rosenberg, J.M. (2023). The design and effects of educational data science workshops for early career researchers. *Journal Formative Design in Learning.* https://doi.org/10.1007/s41686-023-00083-7

Hodge, E. M., Rosenberg, J. M., & López, F. A. (2023). "We Don't Teach Critical Race Theory Here": A sentiment analysis of K-12 school and district social media Statements. *Peabody Journal of Education*, 1-15. https://www.tandfonline.com/doi/full/10.1080/0161956X.2023.2261318

Garner, A. V., & Rosenberg, J. (2023). Utilizing iNaturalist to support place-based learning and data analysis. *Science Scope, 46*(7), 54-60. https://www.nsta.org/science-scope/science-scope-fall-2023/using-inaturalist-support-place-based-learning-and-data

Rosenberg, J. M. (2023). Open and useful? Exploring the science education resources on OER Commons. *Contemporary Issues in Technology and Teacher Education, 23*(3), 490-507. https://citejournal.org/wp-content/uploads/2023/07/v23i3Science2.pdf

Borchers, C., Rosenberg, J. M., & Swartzentruber, R. M. (2023). Facebook post data: a primer for educational research. *Educational Technology Research and Development*, 1-20. https://link.springer.com/article/10.1007/s11423-023-10269-2

Rosenberg, J. M., Beymer, P. N., Phun, V., & Schmidt, J. A. (2023). Using intensive longitudinal methods to quantify the sources of variability for situational engagement in science learning environments. *International Journal of STEM Education, 10*(1), 68. https://link.springer.com/article/10.1186/s40594-023-00449-0

Kubsch, M., Krist, C., & Rosenberg, J. M. (2023). Distributing Epistemic Functions and Tasks - A framework for augmenting human analytic power with machine learning in science education research. *Journal of Research in Science Teaching*. https://onlinelibrary.wiley.com/doi/full/10.1002/tea.21803. *Note.* All authors contributed equally.

- Recipient of the 2023 Research Worth Reading Award for the Research in Artificial Intelligence in Science Education Research Interest Group, National Association for Research in Science Teaching
- Recipient of the 2023 GDCP (German Society for the Teaching of Chemistry and Physics) Best Paper Award

Carpenter, J. P., Morrison, S. A., Rosenberg, J. M., & Hawthorne, K. A. (2023). Using social media in pre-service teacher education: The case of a program-wide twitter hashtag. *Teaching and Teacher Education, 124*, 1-17. https://www.sciencedirect.com/science/article/pii/S0742051X23000240

Rosenberg, J., ^Borchers, C., Stegenga, S. M., ^Burchfield, M. A., Anderson, D., & Fischer, C. (2022). How educational institutions reveal students' personally identifiable information on Facebook. *Learning, Media, & Technology.* https://www.tandfonline.com/doi/full/10.1080/17439884.2022.2140672

Rosenberg, J. M., ^Borchers, C., ^Burchfield, M. A., Anderson, D., Stegenga, S. M., & Fischer, C. (2022). Posts about students on Facebook: A data ethics perspective. *Educational Researcher, 51*(8), 547-550. https://journals.sagepub.com/doi/full/10.3102/0013189X221120538

Rosenberg, J. M., Kubsch, M., Wagenmakers, E.-J., & Dogucu, M. (2022). Making sense of uncertainty in the science classroom: A Bayesian approach. *Science & Education*, 31, 1239–1262. https://link.springer.com/article/10.1007/s11191-022-00341-3

Jones. R. S., & Rosenberg, J. M. (2022). Characterizing whole class discussions about data and statistics with conversation profile analysis. *Journal of Mathematical Behavior, 67*, 1-16. https://www.sciencedirect.com/science/article/abs/pii/S0732312322000645

Rosenberg, J. M., Schultheis, E., Kjelvik, M., Reedy, A., & +Sultana, O. (2022). Big data, big changes? A survey of K-12 science teachers in the United States on which data sources and tools they use in the classroom. *British Journal of Educational Technology, 53*(5), 1179-1201. https://bera-journals.onlinelibrary.wiley.com/doi/10.1111/bjet.13245

+Michela, E., Rosenberg, J., Kimmons, R., +Sultana, O., ^Burchfield, M. A., & ^Thomas, T. (2022). "We are trying to communicate the best we can": Understanding districts' communication on Twitter during the COVID-19 pandemic. *AERA Open, 8*, 1-18. https://doi.org/10.1177/23328584221078542

Trout Fryxell, R. T., Camponovo, M., Smith, B., Butefish, K., Rosenberg, J. M., Andsager, J. L., ... & Willis, M. P. (2022). Development of a community-driven mosquito surveillance program for vectors of La Crosse virus to educate, inform, and empower a community. *Insects, 13*(2), 164. https://www.mdpi.com/2075-4450/13/2/164

Rutherford, T., Duck, K., Rosenberg, J. M., & Patt, R. (2022). Leveraging mathematics software data to understand student learning and motivation during the COVID-19 pandemic. *Journal of Research on Technology in Education, 54*(1), 94-131. https://www.tandfonline.com/doi/full/10.1080/15391523.2021.1920520

Aguilar, S. J., Rosenberg, J., Greenhalgh, S., Fütterer, T., Lishinski, A., & Fischer, C. (2021). A different experience in a different moment? Teachers' social media use before and during the COVID-19 pandemic. *AERA Open, 7*, 1-17. https://journals.sagepub.com/doi/full/10.1177/23328584211063898

+Lawson, M. A., Herrick, I., R., & Rosenberg, J. M. (2021). Better together: Mathematics and science pre-service teachers' sensemaking about STEM. *Educational Technology & Society, 24*(4), 180–192.

Schweinsburg, M., ... Luis, S. (2021). Same data, different conclusions: Radical dispersion in empirical results. *Organizational Behavior and Human Decision Processes, 165*(7), 228-249. https://www.sciencedirect.com/science/article/pii/S0749597821000200 (*Note.* I was an author and contributor to this large-scale, collaborative project.)

Greenhalgh, S. P., Rosenberg, J. M., & Russell, A. (2021). The influence of policy and context on teachers' social media use. *The British Journal of Educational Technology, 52*(5), 2020-2037. https://bera-journals.onlinelibrary.wiley.com/doi/10.1111/bjet.13096?af=R

Frank, K. A., Lin, Q., Maroulis, S., Strassman, A., Xu, R., Rosenberg, J. M., Hayter, C., Mahmoud, R., Kolak, M., Dietz, T., & Zhang, L. (2021). Hypothetical case replacement can be used to quantify the robustness of trial results. *Journal of Clinical Epidemiology, 134*(6), 150-159. https://www.sciencedirect.com/science/article/pii/S0895435621000366 *Note.* I was a scientific programmer for this project.

Rosenberg, J. M., ^Borchers, C., Dyer, E., Anderson, D. J., & Fischer, C. (2021). Advancing new methods for understanding public sentiment about educational reforms: The case of Twitter and the Next Generation Science Standards. *AERA Open, 7*, 1-17. https://journals.sagepub.com/doi/10.1177/23328584211024261

Ranellucci, J., Robinson, K., Rosenberg, J. M., Lee, Y.-K., Roseth, C., & Linnenbrink-Garcia. (2021). Comparing the roles and correlates of emotions in class and during online video lectures in a flipped anatomy classroom. *Contemporary Educational Psychology, 64*(4), 1-15. https://doi.org/10.1016/j.cedpsych.2021.101966

Akcaoglu, M., Rosenberg, J. M., Hodges, C. B., Hilpert, J. (2021). An exploration of factors impacting middle school students' attitudes toward computer programming. *Computers in the Schools, 38*(1), 19-35. https://doi.org/10.1080/07380569.2021.1882209

Rosenberg, J. M., & Krist, C. (2021). Combining machine learning and qualitative methods to elaborate students' ideas about the generality of their model-based explanations. *Journal of Science Education and Technology, 30*(2), 255-267. https://link.springer.com/article/10.1007%2Fs10956-020-09862-4. *Note.* Both authors contributed equally.

Rosenberg, J. M., & Staudt Willet, K. B. (2021). Balancing participants' privacy and open science in the context of COVID-19: A response to Ifenthaler & Schumacher (2016). *Educational Technology Research & Development, 69*(1), 347-351. https://link.springer.com/article/10.1007/s11423-020-09860-8

Harper, F. K., Rosenberg, J. M., ^Comperry, S., ^Howell, K., & ^Womble, S. (2021). #Mathathome during the COVID-19 Pandemic: Exploring and reimagining resources and social supports for parents. *Education Sciences, 11*(2), 60, 1-24. https://www.mdpi.com/2227-7102/11/2/60

Anderson, D. J., Rowley, B., Stegenga, S., Irvin, P. S., & Rosenberg, J. M. (2020). Evaluating content-related validity evidence using a text-based, machine learning procedure. *Educational Measurement: Issues and Practice, 39*(4), 53-64. https://onlinelibrary.wiley.com/doi/abs/10.1111/emip.12314

Rosenberg, J. M., Reid, J., Dyer, E., Koehler, M. J., Fischer, C., & McKenna, T. J. (2020). Idle chatter or compelling conversation? The potential of the social media-based #NGSSchat network as a support for science education reform efforts. *Journal of Research in Science Teaching, 57*(9), 1322-1355. https://onlinelibrary.wiley.com/doi/10.1002/tea.21660

Carpenter, J., Rosenberg, J. M., Dousay, T., Romero-Hall, E., Trust, T., Kessler, A., Phillips, M., Morrison, S., Fischer, C. & Krutka, D. (2020). What should teacher educators know about technology? Perspectives and self-assessments. *Teaching and Teacher Education, 95*(10), 103-124. https://doi.org/10.1016/j.tate.2020.103124

Ranellucci, J., Rosenberg, J. M., & Poitras, E. (2020). Exploring pre-service teachers' use of technology: The technology acceptance model and expectancy-value theory. *Journal of Computer Assisted Learning, 36*(6), 810-824. http://dx.doi.org/10.1111/jcal.12459

Schmidt, J. A., Beymer, P. N., Rosenberg, J. M., Naftzger, N. J., & Shumow, L. (2020). Experiences, activities, and personal characteristics as predictors of engagement in STEM-focused summer programs. *Journal of Research in Science Teaching, 57*(8), 1281-1309. https://onlinelibrary.wiley.com/doi/full/10.1002/tea.21630

Greenhalgh, S. P., Rosenberg, J. M., Koehler, M. J., Akcaoglu, M., & Staudt Willet, K. B. (2020). Identifying multiple learning spaces within a single teacher-focused Twitter hashtag. *Computers & Education, 148*(4). https://doi.org/10.1016/j.compedu.2020.103809

Beymer, P. N., Rosenberg, J. M., & Schmidt, J. A. (2020). Does choice matter or is it all about interest? An investigation using an experience sampling approach in high school science classrooms. *Learning and Individual Differences, 78*(2), 1-15. https://doi.org/10.1016/j.lindif.2019.101812

Rosenberg, J. M., +Edwards, A., & Chen, B. (2020). Getting messy with data: Tools and strategies to help students analyze and interpret complex data sources. *The Science Teacher, 87*(5). https://learningcenter.nsta.org/resource/?id=10.2505/4/tst20_087_05_30

Xu, R., Frank, K. A., Maroulis, S., & Rosenberg, J. M. (2019). konfound: Command to quantify robustness of causal inferences. *The Stata Journal, 19*(3), 523-550. https://journals.sagepub.com/doi/full/10.1177/1536867X19874223

Blondel, D. V., Sansone, A., Rosenberg, J. M., Yang, B. W., Linennbrink-Garcia, L., & Schwarz-Bloom, R. D. (2019). Development of an online experiment platform (Rex) for high school biology. *Journal of Formative Design for Learning, 3*(1) 62-81. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6716597/

Henriksen, D., Mehta, R. & Rosenberg, J. (2019). Supporting a creatively focused technology fluent mindset among educators: survey results from a five-year inquiry into teachers' confidence in using technology. *Journal of Technology and Teacher Education, 27*(1), 63-95. https://www.learntechlib.org/primary/p/184724/

Rosenberg, J. M., & +Lawson, M. J. (2019). An investigation of students' use of a computational science simulation in an online high school physics class. *Education Sciences, 9*(49), 1-19. https://www.mdpi.com/2227-7102/9/1/49

Rosenberg, J. M., Beymer, P. N., Anderson, D. J., & Schmidt, J. A. (2018). tidyLPA: An R package to easily carry out Latent Profile Analysis (LPA) using open-source or commercial software. *Journal of Open Source Software, 3*(30), 978. https://doi.org/10.21105/joss.00978

Greenhalgh, S. P., Staudt Willet, K. B., Rosenberg, J. M., & Koehler, M. J. (2018). Tweet, and we shall find: Using digital methods to locate participants in educational hashtags. *TechTrends, 62*(5), 501-508. https://doi.org/10.1007/s11528-018-0313-6

Beymer, P. N., Rosenberg, J. M., Schmidt, J. A., & Naftzger, N. (2018). Examining relationships between choice, affect, and engagement in out-of-school time STEM programs. *Journal of Youth and Adolescence, 47*(6), 1178-1191. https://doi.org/10.1007/s10964-018-0814-9

Akcaoglu, M., Rosenberg, J. M., Ranellucci, J., & Schwarz, C. V. (2018). Outcomes from a self-generated utility value intervention on fifth and sixth-grade students' value and interest in science. *International Journal of Educational Research, 87*, 67-77. https://www.sciencedirect.com/science/article/pii/S0883035517308492

Schmidt, J. A., Rosenberg, J. M., & Beymer, P. (2018). A person-in-context approach to student engagement in science: Examining learning activities and choice. *Journal of Research in Science Teaching, 55*(1), 19-43. https://dx.doi.org/10.1002/tea.21409 (*Note. This article was recognized as one of the 20 most-downloaded articles in JRST between June, 2016 and June, 2018*)

Rosenberg, J.M., Greenhalgh, S.P., Graves Wolf, L. & Koehler, M.J. (2017). Strategies, use, and impact of social media for supporting teacher community within professional development: The case of one urban STEM program. *Journal of Computers in Mathematics and Science Teaching, 36*(3), 255-267. https://www.learntechlib.org/primary/p/180387/

Koehler, M. J., Greenhalgh, S. P., Rosenberg, M. J., & Keenan, S. (2017). What the tech is going on with digital teaching portfolios? Using the TPACK framework to analyze teachers' technological understanding. *Journal of Technology and Teacher Education, 25*, 31-59. http://www.learntechlib.org/p/173346/

Rosenberg, J. M., Greenhalgh, S. P., Koehler, M. J., Hamilton, E., & Akcaoglu, M. (2016). An investigation of State Educational Twitter Hashtags (SETHs) as affinity spaces. *E-Learning and Digital Media, 13*(1-2), 24-44. http://dx.doi.org/10.1177/2042753016672351

Greenhalgh, S. P., Rosenberg, J. M., & Wolf, L. G. (2016). For all intents and purposes: Twitter as a foundational technology for teachers. *E-Learning and Digital Media, 13*(1-2), 81-98. http://dx.doi.org/10.1177/2042753016672131

Hamilton, E., Rosenberg, J. M., & Akcaoglu, M. (2016). Examining the Substitution Augmentation Modification Redefinition (SAMR) model for technology integration. *Tech Trends, 60*, 433-441. http://dx.doi.org/10.1007/s11528-016-0091-y

Rosenberg, J. M., Terry, C. A., Bell, J., Hiltz, V., & Russo, T. (2016). Design guidelines for graduate program social media use. *Tech Trends, 2*, 167-175. http://dx.doi.org/10.1007/s11528-016-0023-x

Rosenberg, J. M., & Koehler, M. J. (2015). Context and Technological Pedagogical Content Knowledge (TPACK): A systematic review. *Journal of Research on Technology in Education, 47*, 186-210. http://dx.doi.org/10.1080/15391523.2015.1052663

Reports (1)

National Academies of Sciences, Engineering, and Medicine. (2026). *Data and Computing in K-12 Education: Foundational Competencies*. Washington, DC: National Academies Press. https://doi.org/10.17226/29303. *N.b.* I contributed as a committee member.

Commissioned Papers, White Papers, Reports, and Articles in the Popular Press (6)

Bilal, D., Fagan, J., Rius, A., Vogiatzis, K., Liu, C., Dahms, H., Duncan, L., Kim, H., Odoi, A., Rosenberg, J., Singh, V., & Sobes, V. (2023). *AI TENNessee Initiative: Education. AI Tennessee Initiative*. *N.b.* I co-chaired the committee that wrote this white paper with C. Liu.

Rosenberg, J. M., & Jones, R. S. (2022). *K-12 data science learning*. Paper commissioned by the National Academies of Sciences, Engineering, and Medicine for *Foundations of Data Science for Students in Grades K-12: A Workshop*. https://www.nationalacademies.org/event/09-13-2022/docs/DD667E469D0EC5DD91A7D85BC839A9852491A3CF9F15

Rosenberg, J. M. (2021). School posts on Facebook could threaten student privacy. *The Conversation*. https://theconversation.com/school-posts-on-facebook-could-threaten-student-privacy-160248

Rosenberg, J. M. (2020). Open-source authorship of data science in education using R. *R Views*. https://rviews.rstudio.com/2020/07/01/open-source-authorship-of-data-science-in-education-using-r/

Rosenberg, J. M. (2018). Opportunities for engaging students in "data practices" in online science classes. *Michigan Virtual Learning Research Institute Blog: Research, Policy, Innovation & Networks*. https://mvlri.org/blog/opportunities-engaging-students-data-practices-online-science-classes/

Rosenberg, J. M., & Ranellucci, J. (2017). Student motivation in online science courses: A path to spending more time on course and higher achievement. *Michigan Virtual Learning Research Institute Blog: Research, Policy, Innovation & Networks*. https://mvlri.org/blog/student-motivation-in-online-science-courses-a-path-to-spending-more-time-on-course-and-higher-achievement/

Naftzger, N., Schmidt, J. A., Shumow, L., Beymer, P. N., & Rosenberg, J. M. (2019). *Exploring the link between STEM activity leader practice and youth engagement: Findings from the STEM IE study*. Washington, DC: American Institutes for Research. https://www.informalscience.org/exploring-link-between-stem-activity-leader-practice-and-youth-engagement-findings-stem-ie-study

Editor-Reviewed Articles Published in Journals (14)

Kimmons, R., Rosenberg, J. M., Ali, F., & West, R. E. (2026). Trends and topics in educational technology, 2026 edition. *TechTrends*. https://doi.org/10.1007/s11528-026-01193-2

Rosenberg, J. M., Bonvini, M., Bray, A., Gong, R., & Yu, B. (2026). Teaching veridical data science: A panel interview with Matteo Bonvini, Andrew Bray, Ruobin Gong, and Bin Yu.

Journal of Statistics and Data Science Education, 34(2), 127-143. https://doi.org/10.1080/26939169.2026.2632565

Allman, B., Kimmons, R., +Wang, W., +Bao, H., Rosenberg, J. M., & Koehler, M. J. (2024). Trends and topics in educational technology, 2024 edition. *TechTrends.* https://doi.org/10.1007/s11528-024-00950-5

Allman, B., Kimmons, R., Rosenberg, J., & Dash, M. (2023). Trends and Topics in Educational Technology, 2023 Edition. *TechTrends, 67*(3), 583-591. https://link.springer.com/article/10.1007/s11528-023-00840-2

Hodge, E. M., López, F. A., & Rosenberg, J. M. (2022). How to respond to community concerns about critical race theory. *Phi Delta Kappan, 104*(3), 48-53. https://journals.sagepub.com/doi/full/10.1177/00317217221136599

Jiang, S., Lee, V. R., & Rosenberg, J. M. (2022). Data science education across the disciplines: Underexamined opportunities for K-12 innovation. *British Journal of Educational Technology, 53*(5), 1073-1079. https://bera-journals.onlinelibrary.wiley.com/doi/full/10.1111/bjet.13258

Kimmons, R., Rosenberg, J.M. (2022). Trends and topics in educational technology, 2022 Edition. *TechTrends, 66*, 134-140. https://doi.org/10.1007/s11528-022-00713-0

Frank K.A., Lin Q., Maroulis S., Mueller A.S., Xu R., Rosenberg J.M., Hayter C.S., Mahmoud R.A., Kolak M., Dietz T., Zhang L. (2022). Response to "Three Comments on the RIR method". *Journal of Clinical Epidemiology, 146*, 134-140. https://pubmed.ncbi.nlm.nih.gov/35131466/

Kessler, A. M., Likely, R., & Rosenberg, J. (2021). Open for whom? The need to define open science for science education. *Journal of Research in Science Teaching, 58*(10), 1590-1595. https://onlinelibrary.wiley.com/doi/epdf/10.1002/tea.21730

Rosenberg, J. M., ^Burchfield, M. B., ^Borchers, C., ^Gibbons, B., & Anderson, D., and Fischer, C. (2021). Posts on Facebook by schools and districts and the potential risks to students' privacy. *Phi Delta Kappan.* https://kappanonline.org/social-media-students-privacy-facebook-rosenberg/

Kimmons, R., Rosenberg, J., & Allman, B. (2021). Trends in educational technology: What Facebook, Twitter, and Scopus can tell us about current research and practice. *TechTrends, 65*(2), 125-136.

Mehta, R., Henriksen, D., & Rosenberg, J. M. (2019). It's not about the tools. *Educational Leadership, 76*(5), 64-69. https://ascd.org/el/articles/its-not-about-the-tools

Rosenberg, J. M., & Logan, C. W. (2017). Review of the book What's Worth Teaching: Rethinking Curriculum in the Age of Technology, by A. Collins. *Teachers College Record.* http://www.tcrecord.org/Content.asp?ContentID=22173

Phillips, M., Harris, J., Rosenberg, J. M., & Koehler, M. J. (2017). TPCK/TPACK research and development: Past, present, and future directions. *Australasian Journal of Educational Technology.* https://doi.org/10.14742/ajet.3907

Contributions to Edited Volumes (13)

Rosenberg, J., King, A., Swartzentruber, R., & Carpenter, J. (2025). The strengths and drawbacks of measuring self-reported and digitally traced social networks. In *Handbook of Social Network Analysis and Education* (pp. 102-113). Edward Elgar Publishing. https://www.elgaronline.com/edcollchap/book/9781035312191/chapter7.xml

Rosenberg, J. M., Bhidiya, H., & Pritchard, C. (2025). Triangulating computational and qualitative methods to measure scientific uncertainty. In P. Wulff, M. Kubsch, & C. Krist (Eds.), *Applying machine learning in science education research: When, how, and why?* (pp. 313–332). Springer.

Rosenberg, J. (in press). *Comparing self-report and digital trace measures of social networks.* In N. Kolleck (Ed.), *Handbook of social network analysis and education research* (Research Handbooks on Social Networks series). Edward Elgar Publishing.

Dai, T., Rosenberg, J. M., & +Lawson, M. A. (2022). Data representations and visualizations in educational research. In T. L. Good & M. McCaslin (*Eds.*), Educational Psychology Section; D. Fisher (*Ed.*), *Routledge encyclopedia of education.* Taylor & Francis: New York, NY. https://doi.org/10.4324/9781138609877-REE148-1

Rosenberg, J. M., +Lawson, M. A., Anderson, D. J., & Rutherford, T. (2020). Making data science count in and for education. In E. Romero-Hall (Ed.), *Research methods in learning design & technology* (pp. 94-110). Routledge: New York, NY. https://osf.io/preprints/edarxiv/hc2dw/download/

Greenhalgh, S. P., Staudt Willet, B., Rosenberg, J. M., & Koehler, M. J. (2020). Lessons learned from applying Twitter research methods to educational technology phenomena. In E. Romero-Hall (Ed.), *Research methods in learning design & technology* (pp. 64-77). Routledge: New York, NY.

Eidelman, R., Rosenberg, J. M. ,& Shwartz, Y. (2019). Assessing the interaction between self-regulated learning (SRL) profiles and actual learning in the chemistry online blended learning environment (COBLE). In Sampson, D., D. Ifenthaler, M. Spector, P. Isafas, & S. Sergis (Eds), *Learning technologies for transforming teaching, learning and assessment at large scale* (pp. 231-255). Berlin, Germany: Springer.

Herring, M., Koehler, M. J., Mishra, P., Rosenberg, J. M., & Teske, J. (2016). Introduction to the 2nd edition of the TPACK handbook. In M. Herring, M. J. Koehler, & P. Mishra (Eds.), *Handbook of Technological Pedagogical Content Knowledge (TPACK) for educators* (2nd ed., pp. 1-8). New York, NY: Routledge.

Keenan, S., Rosenberg, J. M., Greenhalgh, S. P. & Koehler, M. J. (2016). Examining teachers' technology use through digital portfolios. In L. Liu & D. C. Gibson (Eds.), *Research highlights in technology and teacher education 2016* (pp. 53-60). Chesapeake, VA: Association for the Advancement of Computing in Education.

Phillips, M., Koehler, M. J. & Rosenberg, J. M. (2016). Considering context: Teachers' TPACK development and enactment. In L. Liu & D. C. Gibson (Eds.), *Research highlights in technology and teacher education* (pp. 197-204). Chesapeake, VA: Association for the Advancement of Computing in Education.

Rosenberg, J. M., & Koehler, M. J. (2015). *Context and teaching with technology in the digital age*. In M.L. Niess & H. Gillow-Wiles (Eds.), Handbook of research on teacher education in the digital age (pp. 440-465). Hershey, PA: IGI Global.

Rosenberg, J. M., Greenhalgh, S. P., & Koehler, M. J. (2015). A performance assessment of teachers' TPACK using artifacts from digital portfolios. In L. Liu & D. C. Gibson, *Research highlights in technology and teacher education 2015*. Waynesville, NC: Association for the Advancement of Computing in Education (AACE).

Koehler, M. J., Mishra, P., Akcaoglu, M., & Rosenberg, J. M. (2013). Technological pedagogical content knowledge for teachers and teacher educators. In N. Bharati and S. Mishra (Eds.), *ICT integrated teacher education: A resource book* (pp. 1-8). Commonwealth Educational Media Center for Asia, New Delhi, India.

Papers Published in Refereed Conference Proceedings (34)

Chen, C., Zhao, Y., Rosenberg, J. M., & Hodge, L. L. (2025, August). Mapping the Research Landscape of Culturally Relevant Pedagogy in Computing Education: A Topic Modeling Approach. In *Proceedings of the 2025 ACM Conference on International Computing Education Research* V. 2 (p. 20).

Demirci, S., Dogucu, M., Zieffler, A. & Rosenberg, J. (2024) The Learning Difficulties of Introductory Data Science Students. In E. Jones (Ed) *Proceedings of the International Association for Statistical Education Satellite Conference*. https://www.iase-pub.org/conference_proceedings/IASECP/article/view/254/247

Lishinski, A., Bao, H., & Rosenberg, J. (2024, March). Self-efficacy Interventions for CS1. In *Proceedings of the 55th ACM Technical Symposium on Computer Science Education V.2* (p. 1935). https://dl.acm.org/doi/abs/10.1145/3626253.3635348

Palaguachi, C., Cox, E., Rosenberg, J., Dyer, E., & Krist, C. (2023, October). Automatic speech recognition (ASR) in noisy classrooms: Evaluating the usefulness of three popular ASR tools. In *Learning Sciences Graduate Student Conference*, 2023. https://www.lsgsc.org/_files/ugd/fd445f_74a457de2acb47099455e5a416952899.pdf#page=44

Krist, C., Dyer, E. B., Rosenberg, J., Palaguachi, C., & Cox, E. (2023). Leveraging Computationally Generated Descriptions of Audio Features to Enrich Qualitative Examinations of Sustained Uncertainty. In *Proceedings of the 17th International Conference of the Learning Sciences-ICLS 2023*, pp. 1258-1261. International Society of the Learning Sciences. https://repository.isls.org/handle/1/9894

Thomas, T., Rosenberg, J. M., Penner, E. & Burchfield, M. (2022). Racism and school districts' social media posts: "Is it being talked about?". Proceedings for the *Social Media and Society* conference.

Burchfield, M., Rosenberg, J. M., & Stegenga, S. (2022). Updated or outdated consent? A study of the media release policies of U.S. K-12 school districts. Proceedings for the *Social Media and Society* conference.

Lishinski, A., Narvaiz, S., and Rosenberg, J. M. (2022). Self-efficacy, interest, and belongingness – URM Students' momentary experiences in CS1. *Proceedings of the 2022 ACM Conference on International Computing Education Research* (ICER '22). https://dl.acm.org/doi/abs/10.1145/3501385.3543958. *Note.* Unlike most of academia, premiere conferences in computing education are selective venues for archival research. These conference proceedings exceed many journals in their selectivity, visibility, and impact.

Carpenter, J., Morrison, S., Rosenberg, J., & Hawthorne, K. (2022). Professional learning networks for pre-service teachers: A social network analysis of a program-wide teacher education Twitter hashtag. In *Society for Information Technology & Teacher Education International Conference* (pp. 1489-1497). Association for the Advancement of Computing in Education (AACE). https://www.learntechlib.org/p/220934/

Borchers, C., Rosenberg, J., Gibbons, B., Burchfield, M., & Fischer, C. (2021). To scale or not to scale: Comparing popular sentiment analysis dictionaries on educational Twitter data. In I.-H. Hsiao, S. Sahebi, F. Bouchet, & J.-J. Vie (Eds), *Proceedings of the 14th International Conference on Educational Data Mining* (pp. 619-624). Paris, France. https://educationaldatamining.org/EDM2021/virtual/static/pdf/EDM21_paper_122.pdf

Burchfield, M., Rosenberg, J., Borchers, C., Thomas, T., Gibbons, B., & Fischer, C. (2021). Are violations of student privacy "quick and easy"? Investigating the privacy of students' images and names in the context of K-12 educational institution's posts on Facebook. In I.-H. Hsiao, S. Sahebi, F. Bouchet, & J.-J. Vie (Eds), *Proceedings of the 14th International Conference on Educational Data Mining* (pp. 744-749), Paris, France. https://educationaldatamining.org/EDM2021/virtual/static/pdf/EDM21_paper_276.pdf.

Lishinski, A., & Rosenberg, J. (2021). All the pieces matter: The relationship of momentary self-efficacy and affective experiences with CS1 achievement and interest in computing. *Proceedings of the 17th ACM Conference on International Computing Education Research*, 252–265. https://doi.org/10.1145/3446871.3469740. *Note.* This presentation received the best poster award. Unlike most of academia, premiere conferences in computing education are

selective venues for archival research. These conference proceedings exceed many journals in their selectivity, visibility, and impact.

Rosenberg, J. M., & Kubsch, M. (2021). Considering K-12 learners' use of Bayesian methods. In E. de Vries, Y. Hod, & J., Ahn (Eds.), *The International Society of the Learning Sciences 2020 Conference Proceedings* (pp. 947-948). International Society of the Learning Sciences. https://repository.isls.org/handle/1/7635

Rosenberg, J. M., Galas, E., & Staudt Willet, K.B. (2021). Who are the data scientists in education? An investigation of the identities and work of individuals in diverse roles. In E. de Vries, Y. Hod, & J., Ahn (Eds.), *The International Society of the Learning Sciences 2020 Conference Proceedings* (pp. 1057-1058). International Society of the Learning Sciences. https://repository.isls.org/handle/1/7394

Kubsch, M., Rosenberg. J. M., & Krist, C. (2021). Beyond supervision: Human / machine distributed learning in learning sciences research. In E. de Vries, Y. Hod, & J., Ahn (Eds.), *The International Society of the Learning Sciences 2020 Conference Proceedings* (pp. 897-898). International Society of the Learning Sciences. https://repository.isls.org/handle/1/7609

Rosenberg, J. M., & Nguyen, H. (2021). How K-12 school districts communicated during the COVID-19 pandemic: A study using Facebook data. *Proceedings of the 11th International Conference on Learning Analytics & Knowledge (LAK21).* https://joshuamrosenberg.com/post-prints/LAK21-Rosenberg-Nguyen-Districts-Facebook-published.pdf

Rosenberg, J. M., & Staudt Willet, K. B. (2021). Advancing social influence models in learning analytics. *Companion proceedings of the 11th International Conference on Learning Analytics & Knowledge (LAK21).* http://ceur-ws.org/Vol-2868/article_2.pdf

Lishinski, A., & Rosenberg, J. M. (2021). How CS1 students experienced COVID-19 in the moment: using an experience sampling approach to understand the transition to emergency remote instruction. In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education* (p. 1254). https://dl.acm.org/doi/abs/10.1145/3408877.3439657

+Mann, M., ^Bui, H., ^Gibbons, B., +Lishinski, A., Dyer, E., & Rosenberg, J. M. (2021). "Not my subject"?: A survey of teachers regarding the implementation of new K-8 computing education standards. In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education* (p. 1343). https://dl.acm.org/doi/abs/10.1145/3408877.3439666

Rosenberg, J. M., +Schmidt, A., Rosenberg, A. M., +Longnecker, J., & +Mann M. (2020). Becoming 'tidier' over time. Studying #tidytuesday as a social media-based context for learning to visualize data. In M. Gresalfi and I. Horn (Eds.), *The Interdisciplinarity of the Learning Sciences: The International Conference of the Learning Sciences 2020 Conference Proceedings* (Vol 3., pp. 1811-1812). ISLS. https://repository.isls.org/handle/1/6455

Jones R. S., & Rosenberg, J. M. (2020). Studying whole class discussions at scale. In M. Gresalfi and I. Horn (Eds.), *The Interdisciplinarity of the Learning Sciences: The International*

Conference of the Learning Sciences 2020 Conference Proceedings (Vol 5., pp. 2499-2506). ISLS. https://repository.isls.org/handle/1/6612

D'Angelo, C., Dyer, E. B., Krist, C., Rosenberg, J. M., & Bosch, N. (2020). Advancing computational grounded theory for audiovisual data from mathematics classrooms. In M. Gresalfi and I. Horn (Eds.), *The Interdisciplinarity of the Learning Sciences: The International Conference of the Learning Sciences 2020 Conference Proceedings* (Vol 4., pp. 2393-2394). ISLS.

Dyer, E. B., D'Angelo, D., Bosch, N., Krist, C., & Rosenberg, J. M. (2020). Analyzing learning with speech analytics and computer vision methods: Technologies, principles, and ethics. In M. Gresalfi and I. Horn (Eds.), *The Interdisciplinarity of the Learning Sciences: The International Conference of the Learning Sciences 2020 Conference Proceedings* (Vol 5., pp. 2651-2652). ISLS. https://joshuamrosenberg.com/publications/dyer-et-al-icls-2020.pdf

Rosenberg, J. M. (2020). More confidently uncertain? Teaching learners to apply Bayesian methods to make sense of scientific phenomena. In M. Gresalfi and I. Horn (Eds.), *The Interdisciplinarity of the Learning Sciences: The International Conference of the Learning Sciences 2020 Conference Proceedings* (Vol 5., pp. 2681-2682). ISLS.

Lishinski, A., & Rosenberg, J. (2020, February). Accruing interest: What experiences contribute to students developing a sustained interest in computer science over time? In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education* (p. 1414). https://dl.acm.org/doi/abs/10.1145/3328778.3372568

Rosenberg, J., & Lishinski, A. (2020, February). Variable interest rate: What experiences explain differences in interest in computer science among students? In *Proceedings of the 51st ACM Technical Symposium on Computer Science Education* (p. 1404). https://dl.acm.org/doi/abs/10.1145/3328778.3372537

Carpenter, J., Rosenberg, J. M., Dousay, T., Romero-Hall, E., Trust, T., Kessler, A., Phillips, M., Morrison, S., Fischer, C. & Krutka, D. (2019). What do teacher educators think of teacher education technology competencies?. In K. Graziano (Ed.), *Proceedings of Society for Information Technology & Teacher Education International Conference* (pp. 796-801). Las Vegas, NV, United States: Association for the Advancement of Computing in Education (AACE). Retrieved April 18, 2019 from https://www.learntechlib.org/primary/p/207735/.

Peterson, A., Freer, D., & Rosenberg, J. M. (2017). Interacting with purpose: What is the difference between face-to-face and online student relationships in a combined program? In *Proceedings of Society for Information Technology & Teacher Education International Conference* (pp. 3411-3414). Austin, TX: Association for the Advancement of Computing in Education. Retrieved from https://www.learntechlib.org/p/177955/

Krist, C., & Rosenberg, J. M. (2016). Finding patterns in and refining characterizations of students' epistemic cognition: A computational approach. In Looi, C.-K., Polman, J., Cress, U., & Reimann, P. (Eds.), *Transforming Learning, Empowering Learners: The International*

Conference of the Learning Sciences Proceedings 2016 (Vol. 2, pp. 1223-1224). Singapore, Singapore: ICLS.

Rosenberg, J. M., Koehler, M. J., Akcaoglu, M., Greenhalgh, S. P. & Hamilton, E. (2016). State Educational Twitter Hashtags: An introduction and research agenda. In *Proceedings of Society for Information Technology & Teacher Education International Conference 2016* (pp. 355-360). Chesapeake, VA: Association for the Advancement of Computing in Education. Retrieved from http://www.editlib.org/p/171698

Greenhalgh, S. P., Rosenberg, J. M. & Wolf, L. G. (2016). For every tweet there is a purpose: Twitter within (and beyond) an online graduate program. In *Proceedings of Society for Information Technology & Teacher Education International Conference 2016* (pp. 2044-2049). Chesapeake, VA: Association for the Advancement of Computing in Education (AACE). Retrieved from http://www.editlib.org/p/171972

Schwarz, C. V., Ke, L., Lee, M, & Rosenberg, J. M. (2014). Developing mechanistic explanations of phenomena: Case studies of two fifth grade students' epistemologies in practice over time. In J. L. Polman, E. A. Kyza, K. O'Neill, I. Tabak, W. R. Penuel, A. S. Jurow, . . . L. D'Amico (Eds.), *Learning and becoming in practice: The international conference of the learning sciences (ICLS) 2014* (Vol. 1, pp. 182-189). Boulder, CO: ISLS. http://www.isls.org/icls2014/Proceedings.html

Rosenberg, J. M., & Koehler, M. (2014). Context and Technological Pedagogical Content Knowledge: A content analysis. In M. Searson & M. Ochoa (Eds.), *Proceedings of Society for Information Technology & Teacher Education International Conference 2014* (pp. 2412-2417). Chesapeake, VA: AACE. Retrieved from http://www.editlib.org/p/131183

Greenhalgh, S. P., Rosenberg, J. M., Zellner, A. & Koehler, M. J. (2014). Zen and the art of portfolio maintenance: Best practices in course design for supporting long-lasting portfolios. In M. Searson & M. Ochoa (Eds.), *Proceedings of Society for Information Technology & Teacher Education International Conference 2014* (pp. 1604-1610). Chesapeake, VA: AACE. Retrieved from http://www.editlib.org/p/131027

Rosenberg, J., Terry, C., Bell, J., Hiltz, V., Russo, T. & The EPET Social Media Council (2014). What we've got here is failure to communicate: Social media best practices for graduate school programs. In M. Searson & M. Ochoa (Eds.), *Proceedings of Society for Information Technology & Teacher Education International Conference 2014* (pp. 1210-1215). Chesapeake, VA: AACE. Retrieved from http://www.editlib.org/p/130949

Rosenberg, J. (2013). Review of mobile device use policies in public high schools. In R. McBride & M. Searson (Eds.), *Proceedings of Society for Information Technology & Teacher Education International Conference 2013* (pp. 3774-3779). Chesapeake, VA: AACE. Retrieved from http://www.editlib.org/p/48698

Research Software

Rosenberg, J. M., van Lissa, C. J., Beymer, P. N., Anderson, D. J., Schell, M. J. & Schmidt, J. A. (2019). *tidyLPA: Easily carry out Latent Profile Analysis (LPA) using open-source or commercial software* [R package]. https://data-edu.github.io/tidyLPA/

Rosenberg, J. M., Xu, R., & Frank, K. A. (2019). *konfound: Quantify the robustness of causal inferences* [R package]. https://jrosen48.github.io/konfound/

Rosenberg, J. M., Xu, R., & Frank, K. A. (2019). *Konfound-It!: Quantify the robustness of causal inferences* [R Shiny application]. http://konfound-it.com.

Estrellado, R. A., Bovee, E. A., Mostipak, J., Rosenberg, J. M., & Velásquez, I. C. (2019). *dataedu: Package for Data Science in Education Using R.* https://github.com/data-edu/dataedu

Rosenberg, J. M. (2020). *tidykids: State-by-State Spending on Kids Dataset.* https://jrosen48.github.io/tidykids/

Lishinski, A., & Rosenberg, J. M. (2019). *Short message survey: An open-source, text-message based application for the experience sampling method.* https://github.com/picsul/short-message-survey

Rosenberg, J. M. (2016). Diffusion & temperature. *Lab Interactive Simulation.* https://lab.concord.org/interactives.html#interactives/external-projects/msu/temperature-diffusion.json

Presentations

Information on all presentations are provided here; I often present at AERA, ISLS, and NARST, among others.

Invited Talks

Rosenberg, J. M. (May, 2026). *Marrying education data science and data science education* [Keynote talk]. Educational Data Science Conference, Stanford, CA. https://edsconference.stanford.edu/

Rosenberg, J. M. (October, 2025). *Public opinion about science education: Social media as a context and data source* [Invited talk]. University of Memphis.

Rosenberg, J. M. (October, 2024). *The tensions and opportunities surrounding emerging educational technologies* [Keynote address]. International Computer & Instructional Technologies Symposium (ICITS), Kastamonu University, Turkey. https://icits2024.kastamonu.edu.tr

Rosenberg, J. M. (April, 2021). *AI and ML and data! Oh my! Supporting teachers' and learners' work by considering the human sides of data science.* Keynote presentation at the LEAD Graduate School and Research Network retreat. The University of Tübingen, Baden-Württemberg, Germany.

Rosenberg, J. M. (October, 2021). *All together now: Leveraging data science techniques alongside traditional approaches to understand learning.* Invited presentation at the International Conference on Education Research. Seoul National University, Seoul, South Korea.

Rosenberg, J. M. (February, 2020). *Studying education-focused Twitter hashtags in light of state-based and national policies and practices.* Presentation at the 2020 Spring Seminar Series at the Martin School of Public Policy at the University of Kentucky, Lexington, KY.

Rosenberg, J.M. (September, 2019). *Making data science education count: Exploring the integration of data science into education.* Presentation at the Middle Tennessee State University Mathematics and Science Education Doctoral Seminar series. Middle Tennessee State University, Murfreesboro, TN.

Rosenberg, J. M. (February, 2019). *Making sense of recent advances in the Technological Pedagogical Content Knowledge framework.* English International Congress at the Universidad Técnica del Norte, Ibarra, Ecuador.

Select Workshops

I have led many workshops; additional references are provided in the above link.

Rosenberg, J. M. & D'Angelo, C. (2022, March). *An Introduction to Data Science in Science Education Using R.* National Association for Research in Science Teaching. *Note.* This session was sponsored by the Contemporary Methods Research Interest Group (RIG).

Kellogg, S., Jiang, S., Moore, R., & Rosenberg, J. M. (2021, August). *A LASER Focus on Understanding and Improving STEM Education.* Partnerships for Expanding STEM Education Research in STEM (AERA & ICPSR). https://github.com/laser-institute/aera-workshop

Rosenberg, J. M. (2021, July). *An Introduction to Natural Language Processing in Science Education.* Workshop carried out at the Machine Learning and Computer-Based Text Analysis conference, Kiel, Germany. https://joshuamrosenberg.com/post/2021/07/19/an-introduction-to-natural-language-processing-in-science-education/

Select Outreach and Community Engagement

Rosenberg, J. M. (2021, August). *Tools and Strategies to Work with Data in the Science Classroom.* Knox County Schools District Learning Day. https://bit.ly/kcs-dld

D'Angelo, C., & Rosenberg, J. M. (2021, April). *Analyzing Education Data with Open Science Best Practices, R, and OSF*. Webinar through the Center for Open Science. https://www.youtube.com/watch?v=WxdWzTIzYmI

Teaching

Courses Taught

Instructor at the University of Tennessee, Knoxville:

- *Foundations of Educational Data Science I* (STEM 680, M.A. and Ph.D. class)
- *Foundations of Educational Data Science II* (STEM 685, M.A. and Ph.D. class)
- *Visualizing Data Using R* (STEM 691, M.A. and Ph.D. class)
- *Capstone in Educational Data Science* (STEM 695, M.A. and Ph.D. class)
- *STEM Education Seminar* (STEM 612, Ph.D. class)
- *Trans-Departmental Seminar II* (TPTE 605, Ph.D. class)
- *Nature of Mathematics and Science Education* (SCED 572, M.A. and Ph.D. class)
- *Teaching Science in Grades 7-12* (TPTE 495, SCED 496, & SCED 543, B.S. & M.A. class)
- *VolsTeach Step 1 and Step 2* (TPTE 110 & TPTE 120, undergraduate-level class)

Instructor at Michigan State University:

- *Psychology of Learning in School and Other Settings* (CEP 800, M.A. class)
- *Approaches to Educational Research* (CEP 822, M.A. class)
- *Technology and Leadership* (CEP 815, M.A. class)

National Leadership and Editorial Roles

Committee Member, Developing Competencies for the Future of Data and Computing: The Role of K-12, National Academies of Sciences, Engineering, and Medicine, 2024-2025

Associate Editor, *Journal of Research on Science Teaching*, 2024-present

Associate Editor, *Journal of Statistics and Data Science Education*, 2023-present

Service

Editorial Service

Editorial Review Board Member, *Review of Educational Research*, 2022-present

Editorial Review Board Member, *Journal of Science Education and Technology*, 2023-present

Editorial Review Board Member, *Journal of Research on Technology in Education*, 2016-2024

Editorial Review Board Member, *Journal of Research in Science Teaching*, 2019-2022

Editorial Review Board Member, *Contemporary Issues in Technology and Teacher Education (Science Education Section)*, 2019-2022

Special Section Editor, *British Journal of Educational Technology*, 2022

Special Issue Editor, *Australasian Journal of Educational Technology*, 2017

Service to the Profession

External Reviewer, Application for Promotion to Associate Professor, Ohio State University, 2024

External Reviewer, Application for Promotion to Associate Professor, Texas Tech, 2024

External Reviewer, Application for Promotion to Professor, Elon University, 2021

Member, OpenSciEd Research Agenda (Assessment) working group, 2021

Review Panels

Panelist, CAREER program, National Science Foundation, *n.d.*

Ad-Hoc Reviewer, Building Capacity in STEM Education Research (one proposal), National Science Foundation, *n.d.*

Ad-Hoc Reviewer, Discovery Research K-12 (three proposals), National Science Foundation, *n.d.*

Ad-Hoc Reviewer, Advancing Informal STEM Learning, National Science Foundation (one proposal), *n.d.*

Panelist, Innovative Technology Experiences for Students and Teachers (ITEST), National Science Foundation, *n.d.* (two panels)

Panelist, Building Capacity in STEM Education Research (BCSER), National Science Foundation, *n.d.*

Panelist, Discovery Research PreK-12 (DRK-12), National Science Foundation, *n.d.*

Service to the Community

Mentor, TN Promise (2022-2023; mentor to 12 college-going students)

Mentor, Diversity in Learning Analytics and Leadership program, https://www.diversityindataandleadershipprog com/

Reviewer, Proposals from Knox County Schools students for the NASA Student Spaceflight Experiment program

Select Ad-Hoc Journal Article Reviewing

- AERA Open
- British Journal of Educational Technology
- Cognition & Instruction
- Computers & Education
- Contemporary Educational Psychology
- Educational Researcher
- Educational Psychologist
- Educational Technology Research & Development
- Journal of the Learning Sciences
- Journal of Open Source Software
- Science Education
- TechTrends
- Urban Education

University Service

2023-2025, Faculty Lead, Data Science Curriculum Committee, College of Emerging and Collaborative Studies, UTK

2022-2023, AI Visioning Working Group (Co-Chair, Education Committee), AI Tennessee Initiative, UTK

2021-2023, Data Science Faculty Committee member, UTK

College-Level Service

2025, Committee Member, Search for Assistant Dean for Research and Engagement, *College of Emerging and Collaborative Studies*, UTK

2023-2024, Search Chair, Unit Head, *Department of Theory and Practice in Teacher Education*, UTK

2023-2024, Search Co-Chair, Assistant/Associate Professor of Educational Data Science, *Department of Theory and Practice in Teacher Education*, UTK

2021-2022, Search Committee Member, Two Tenure Track positions in Learning Design and Technology and Instructional Technology, *Department of Educational Psychology and Counseling*, UTK

Departmental Service

2024-2025, Bylaws Revision Ad-hoc Committee, TPTE, UTK

2022-present, Program Coordinator (Graduate Certificate and MS), Educational Data Science, TPTE, UTK

2022-present, Mentor (five faculty members)

2021-present, Institutional Review Board Departmental Representative (Quantitative), TPTE, UTK

Advising and Student Committees

University of Tennessee, Knoxville

Advisor for Doctoral students:

- 2025-present: Gustavo Alvarez-Suchini (co-advisor with E. Romero-Hall)
- 2025-present: Ronald Rupard
- 2024-present: Bradley Holtz (co-advisor with L. Caudle)
- 2023-present: Cody Pritchard
- 2023-present: Wei Wang
- 2023-present: Hanhui Bao
- 2021-present: Emily McDonald
- 2021-2024: Amanda Garner (postdoctoral scholar at Vanderbilt)
- 2020-present: Omiya Sultana
- 2019-2021: Jennifer Longnecker (PD specialist, DreamBox Learning)

Committee Member for Doctoral students:

- 2025-present: Richard Amoako (Educational Leadership & Policy Studies; ELPS)
- 2022-2024: Amanda Zeller (TPTE)
- 2021-2025: Donna Wortham (ELPS)
- 2021-2024: John Mooneyham
- 2021-2023: Anna Sintsova Banks (Department of Educational Psychology & Counseling)
- 2020-2023: Sarah Narvaiz (Department of Educational Psychology & Counseling)
- 2020-2022: Anthony Schmidt (Department of Educational Psychology & Counseling)
- 2019-2023: Maryrose Weatherton (Department of Ecology and Evolutionary Biology, UTK)
- 2019-2021: Matthew Hensley
- 2018-2020: Shande King

Visiting Scholars Hosted

I have hosted several guest scholars, including Gregor Benz (Technical University of Munich, Germany) and Matthias Fischer (Heidelberg University, Germany).

Invited Participation in Workshops and Working Groups

Invited Participant, *Scoping Workshop on Future of Science Education Research in Times of Artificial Intelligence*, Hannover, Germany, 2025

Invited Participant, *Advancing Critical Social Network Analysis to Address Issues of Equity in Education Policy and Politics*, Spencer Foundation, 2023

Invited Participant, *National Academy of Science - National Science Foundation Working Group - Tools & Transfer*, Data Science for Everyone, 2023

Invited Participant, *Foundations of data science for students in grades K-12: A workshop*, National Academies of Sciences, Engineering and Medicine, 2022

Invited Participant, *Open Scholarship in Education small expert meeting*, Center for Open Science, 2022

Invited Participant, *New applications of social network analysis to education policy: Building the capacity of the field*, Spencer Foundation, 2021

Invited Participant, *Data4Kids: Virtually teaching kids about data science*, Urban Institute, 2021

Competitive Research Training

Early Career Workshop, International Conference of the Learning Sciences, 2020

New Faculty Mentoring Program, AERA Division C, 2019

Media Coverage

Outsmart ChatGPT: 8 Tips for Creating Assignments It Can't Do, *Education Week*, February 14, 2023

Can Digital Tools Detect ChatGPT-Inspired Cheating?, *Education Week*, January 27, 2023

When Does Posting Photos of Students Become a Data Privacy Problem?, *EdSurge*, January 9, 2023

How School Social Media Accounts Put Student Privacy at Risk, *Tech & Learning*, November 30, 2022

Professors publish playbook for how teachers can address parents concerned about CRT, *Fox News*, November 13, 2022

Study: School Facebook Pages May Compromise Student Privacy, *Government Technology*, November 10, 2022

Study Shows Millions of Student Privacy Breaches on Social Media THE Journal, November 7, 2022

School Facebook Pages and Privacy Concerns: What Educators Need to Know *Education Week*, November 2, 2022

Study: Schools' Facebook posts may violate student privacy *K-12 Dive*, November 2, 2022

Study: Schools' Social Media Posts May Be Compromising Student Privacy, *American Educational Research Association*, November 2, 2022

The Tricky Ethics of Being a Teacher on TikTok, *Wired*, September 6, 2022

Teaching Students to Understand the Uncertainties of Science Could Help Build Public Trust, *Education Week*

School Posts on Facebook Could Threaten Student Privacy, *Tennessee News Service*

Teaching R to 7th Graders, *Flowing Data*

Professional Affiliations

- American Educational Research Association
- International Society of the Learning Sciences
- National Association for Research in Science Teaching
- Society for Learning Analytics Research